

Yung Hsing (Winston) Huang

Research Scientist / Bioinformatician



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Houston, TX



SUMMARY/OBJECTIVE

- 7 years of postdoctoral academic and industrial research experience
- 3 years of bioinformatics experience
- Research expertise: Tumor microenvironment, spatial transcriptomics, single-cell multiomics, novel biomarkers/gene signatures
- Lawful permanent resident in United States and a lawful citizen in Canada

WORK EXPERIENCE



Bioinformatician

Houston Methodist Research Institute | Houston, TX, USA | 02/2024 – Present

Spatial Omics Core and Chan Lab

- Integrating **Xenium spatial transcriptomics** and single-cell **T-cell repertoire sequencing** datasets from patients in a phase II clinical trial (Manuscript in preparation)
- Architected end-to-end computational workflows, utilizing SLURM clusters to process large-scale Xenium datasets ($> 10^6$ cells).
- Developing a customized R/Shiny web application, **HMRIBladderR**, for gene/signature prognostic assessment in public human bladder cancer patient datasets.
- Harnessing in-house **single-cell RNAseq** and **Xenium 5k** spatial transcriptomics analysis to identify heterogeneity of cancer-associated fibroblasts (CAF) in non-muscle-invasive and muscle-invasive bladder cancer patients from multiple public datasets ([Gao et al., Sci Adv, 2025](#); and a manuscript in preparation)
- Development of Caspase-1 driven pyroptosis gene signature using Cox proportional hazard (CPH) model and utilizing in-house **Xenium spatial transcriptomics** to visualize the intratumoral pyroptotic niche that results in local increase of lymphocyte-to-neutrophil ratio ([Wong et al., Nat Comm, 2026](#))
- Leveraging **Xenium spatial transcriptomics** to characterize the niche composition and essential ligand-receptor interactions of therapy-induced senescent tumor cells that are enriched in minimal residual cancers (Liu et al., *Nat Comm*, under revision).
- Received Exceptional Employee Bonus in 2025 and 2026 for top 10% performing employees at Houston Methodist Hospital
- Mentoring two PhD students for the bioinformatics aspect of their research projects.

Bioengineer

3D Systems | Houston, TX, USA | 06/2022 – 11/2023

I worked for a subsidiary company of 3D Systems, Systemic Bio, which focused on development of organ/tumor-on-chip development for the purposes of preclinical drug efficacy/toxicity assessment. My main job functions included:

- Assay development and optimization for samples collected from 3D organ/tumor-on-chip systems including viability/cytotoxicity assays (LDH, Cell-Titer Glo, MTS, CCK8), histology assays (cryosectioning of 3D-printed hydrogels, IHC/H&E staining), analyte quantification assays (ELISA of soluble proteins, bile acids, enzyme activity colorimetric assays) and nucleic acid assays (qPCR, PicoGreen).
Assessment of biocompatibility of organ/tumor-on-chip systems using primary human hepatocytes and immortalized human cell lines using various biomaterials, microfluidic systems and hydrogel print patterns
- Establishment of pipelines for biological/biochemical assay data collection, analyses, visualization and interpretation
Maintenance of core experimental equipment including Leica Cryostat, MiniAmp™ Plus Thermal Cycler, Nanodrop, QuanStudio 6, MACSQuant 10 Analyzer Flow Cytometer and Promega Glomax Plate Reader
- Participation of partnership million-scale funding projects using organ/tumor-on-chip for preclinical drug screening with two major pharmaceutical companies.

Postdoctoral Fellow

Houston Methodist Research Institute | Houston, TX, USA | 04/2019 – 05/2022

- Identified an innovative therapeutic target NHE6 for improved therapy efficacy in resistant multiple myeloma patients ([Huang et al., Drug Res Updates, 2026](#))
- Writing and preliminary data generation for a successfully funded CPRIT grant titled "[Targeting NHE6 to improve clinical efficacy of daratumumab in myeloma](#)" (Grant ID: RP220639)

**Master of Science (M.Sc.) in Biomedical Informatics****University of Texas Health Science Center at Houston | Houston, TX, USA | Class of 2024**

- Practicum project on in silico drug screening for KMT2A::AFF1 fusion protein ([Huang et al., iScience, 2026](#))

Doctor of Philosophy (Ph.D.) in Lab Medicine and Pathology**University of Alberta | Edmonton, AB, Canada | Class of 2019**

- Development of a 3D cell culture model for therapy assessment and molecular analyses of primary myeloma cells and myeloma cell lines ([Huang et al., Cancers, 2018](#); [Huang et al., Cancers, 2021](#))
- Development and preclinical assessment of anti-CD38-conjugated nanoparticles encapsulating STAT3 inhibitor as an innovative therapeutic agent for multiple myeloma ([Huang et al., Cancers, 2019](#))

Bachelor of Science, Honours Degree (BSc.(Hon.)) in Biochemistry**University of British Columbia | Vancouver, BC, Canada****PATENT**

Pyroptosis-Associated Gene Signature (CASCADE) for Predicting Treatment Response in Bladder Cancer – Filed internally at Houston Methodist Research Institute # DISC2025014

**RELEVANT CORE SKILLS**

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|---|--|
| ➤ In silico drug screening and docking | ➤ Gene signature development |
| ➤ Public dataset mining (TCGA, GEO, EMBL-EBI etc) | ➤ NGS analyses (bulkRNA, scRNA, spatial, TCR, riboseq) |
| ➤ Version control and script management (GitHub) | ➤ Python and R |
| ➤ Cloud computation system (SLURM, HPC) | ➤ Linux for large-scale data manipulation |
| ➤ In vitro/vivo molecular biology techniques | ➤ Knowledge on wet lab techniques |
| ➤ Automation with Local LLM model | |

**RESEARCH CONTRIBUTIONS****Peer-reviewed publications**

- Chen, L. *, Cheng, F. *, **Huang, Y.-H.***, Wang, Y. *, He, J., Liu, J., Wang, Z., Lin, P., Fan, Y., Bao, L., Yi, Q., & Yang, J. (2026). NHE6-driven endosome-autophagy axis confers proteasome inhibitor resistance in multiple myeloma. *Drug Resistance Updates*. <https://doi.org/10.1016/j.drug.2026.101436>
- Kumar, H. *, Tsang, S. V. *, Lee, M.-S. *, **Huang, Y.-H.***, Choi, J., Lotlikar, S. R., Song, J.-Y., Toh, A., Choi, Y.-L., Brannan, K. W., & Kim, P. (2026). FusionTarget: Computational framework for drug repurposing against modeled fusion protein structures from genomic breakpoints. *iScience*. <https://doi.org/10.1016/j.isci.2026.116076>
- Wong, S. Q. *, Hayashi, K. *, Subel, E. *, Gao, H., **Huang, Y. H.**, Nikolo, F., Karabicici, M., Hoi, X. P., Alonzo, M., Kasabyan, A., Tsouko, E., Bouchier-Hayes, L., Shin, C., Porter, S., Garcia, H., Yin, Z., Wong, S. T., El-Zaatari, Z., Satkunasivam, R., Chen, C., Brooks, M., Zhou, L., Korentzelos, D., Gong, Y., Li, Y., Titus, R. & Chan, KS. (2026). Chemotherapy-induced caspase-1 dependent pyroptosis in cancer cells remotely skews myelopoiesis to drive pro-tumorigenic systemic neutrophil-dominant inflammation. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-71471-3>
- Gao, H., Wong, S., Subel, E., **Huang, Y. H.**, Lee Y. C., Hayashi, K., Alonzo, M. E., Karabicici, M., Hoi, X. P., Kasabyan, A., Mo, Q., Zaatari, Z., Shen, S., Satkunasivam, R., Nikolos, F. & Chan, K. S. (2025). Caspase-1 dependent pyroptosis converts αSMA+ CAFs into collagen-IIIhigh iCAFs to fuel chemoresistant cancer stem cells. *Sci. Adv.* 11, eadt8697(2025). <https://doi.org/10.1126/sciadv.adt8697>
- Liu, H., He, J., Bagheri-Yarmand, R., Li, Z., Liu, R., Wang, Z., Bach, D.-H., **Huang, Y. H.**, Lin, P., Guise, T. A., Gagel, R. F., & Yang, J. (2022). Osteocyte ClITA aggravates osteolytic bone lesions in myeloma. *Nature Communications*, 13(1), 3684. <https://doi.org/10.1038/s41467-022-31356-7>
- Wang, Q., Lin, Z., Wang, Z., Ye, L., Xian, M., Xiao, L., Su, P., Bi, E., **Huang, Y.-H.**, Qian, J., Liu, L., Ma, X., Yang, M., Xiong, W., Zu, Y., Pingali, S. R., Xu, B., & Yi, Q. (2022). RARγ activation sensitizes human myeloma cells to carfilzomib treatment through the OAS-RNase L innate immune pathway. *Blood*. <https://doi.org/10.1182/blood.2020009856>
- Wang, Z., He, J., Bach, D.-h., **Huang, Y.-H.**, Li, Z., Liu, H., Lin, P., & Yang, J. (2022). Induction of m⁶A methylation in adipocyte exosomal LncRNAs mediates myeloma drug resistance. *Journal of Experimental & Clinical Cancer Research*. <https://doi.org/10.1186/s13046-021-02209-w>
- Huang, Y.-H.**, Almowaled, M., Li, J., Venner, C., Sandhu, I., Peters, A., Lavasanifar, A., & Lai, R. (2021). Three-dimensional reconstructed bone marrow matrix culture improves the viability of primary myeloma cells in-vitro via a STAT3-dependent mechanism. *Current Issues in Molecular Biology*. <https://www.mdpi.com/1467-3045/43/1/26>

9. Saqr, A., Vakili, M. R., **Huang, Y.-H.**, Lai, R., & Lavasanifar, A. (2019). Development of traceable rituximab-modified PEO-polyester micelles by postinsertion of PEG-phospholipids for targeting of B-cell lymphoma. *ACS Omega*. <https://doi.org/10.1021/acsomega.9b02910>
10. Haque, M., Li, J., **Huang, Y.-H.**, Almowaled, M., Barger, C. J., Karpf, A. R., Wang, P., Chen, W., Turner, S. D., & Lai, R. (2019). NPM-ALK is a key regulator of the oncoprotein FOXM1 in ALK-positive anaplastic large cell lymphoma. *Cancers*. <https://doi.org/10.3390/cancers11081119>
11. **Huang, Y. H.**, Vakili, M. R., Molavi, O., Morrissey, Y., Wu, C., Paiva, I., Soleimani, A. H., Sanaee, F., Lavasanifar, A., & Lai, R. (2019). Decoration of anti-CD38 on nanoparticles carrying a STAT3 inhibitor can improve the therapeutic efficacy against myeloma. *Cancers*, 11(2). <https://doi.org/10.3390/cancers11020248>
12. **Huang, Y.-H.**, Molavi, O., Alshareef, A., Haque, M., Wang, Q., Chu, M. P., Venner, C. P., Sandhu, I., Peters, A. C., Lavasanifar, A., & Lai, R. (2018). Constitutive activation of STAT3 in myeloma cells cultured in a three-dimensional, reconstructed bone marrow model. *Cancers*. <https://doi.org/10.3390/cancers10060206>
13. Wu, C., Gupta, N., **Huang, Y. H.**, Zhang, H. F., Alshareef, A., Chow, A., & Lai, R. (2018). Oxidative stress enhances tumorigenicity and stem-like features via the activation of the Wnt/ β -catenin/MYC/Sox2 axis in ALK-positive anaplastic large-cell lymphoma. *BMC Cancer*, 18(1), 1–11. <https://doi.org/10.1186/s12885-018-4300-2>
14. Soleimani, A. H., Garg, S. M., Paiva, I. M., Vakili, M. R., Alshareef, A., **Huang, Y. H.**, Molavi, O., Lai, R., & Lavasanifar, A. (2017). Micellar nano-carriers for the delivery of STAT3 dimerization inhibitors to melanoma. *Drug Delivery and Translational Research*, 7(4), 571–581. <https://doi.org/10.1007/s13346-017-0369-4>
15. Alshareef, A., Zhang, H. F., **Huang, Y. H.**, Wu, C., Zhang, J. D., Wang, P., El-Sehemy, A., Fares, M., & Lai, R. (2016). The use of cellular thermal shift assay (CETSA) to study Crizotinib resistance in ALK-expressing human cancers. *Scientific Reports*, 6(1), 1–12. <https://doi.org/10.1038/srep33710>
16. Wu, C., Zhang, H. F., Gupta, N., Alshareef, A., Wang, Q., **Huang, Y. H.**, Lewis, J. T., Douglas, D. N., Kneteman, N. M., & Lai, R. (2016). A positive feedback loop involving the Wnt/ β -catenin/MYC/Sox2 axis defines a highly tumorigenic cell subpopulation in ALK-positive anaplastic large cell lymphoma. *Journal of Hematology and Oncology*, 9(1), 1–14. <https://doi.org/10.1186/s13045-016-0349-z>